AI Recipe Generator

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***Abstract*—Cooking can often feel overwhelming when deciding what meal to prepare with the ingredients already available. This study introduces an AI-based Recipe Generation and Recommendation System that merges Natural Language Processing (NLP) and computer vision to assist users in creating customized recipes within seconds. The system accepts textual and visual inputs—such as typed ingredients or an uploaded image—and produces detailed recipes tailored to dietary goals, cuisine preferences, and health restrictions. Two versions of the model were developed: a rule-based baseline that matches ingredients by overlap, and an advanced hybrid model that combines YOLOv5-based ingredient detection, a FastAPI backend, and a Streamlit interface for interactive visualization. The enhanced system significantly improves recipe accuracy, personalization, and user engagement while reducing food wastage.**

Artificial Intelligence, NLP, Computer Vision, Recipe Generation, FastAPI, Streamlit, YOLOv5, Smart Kitchen

I. Introduction

In recent years, the growing integration of Artificial Intelligence into daily life has transformed how people approach routine tasks. Cooking, a fundamental human activity, has not been left behind. However, in today's fast-paced environment, people often struggle to decide what to cook using the ingredients they already have. Browsing through online recipe databases or manually searching for ideas consumes valuable time and can feel repetitive and overwhelming.

This project aims to address that issue by developing an AI-powered system that can understand both natural language and visual data to provide intelligent, customized recipe recommendations. Unlike conventional systems that rely on static keyword matching, this approach uses Natural Language Processing (NLP) for contextual understanding and Computer Vision (YOLOv5) for ingredient detection. Together, these technologies make the system capable of interpreting human input in multiple forms — whether through typed text or images.

The system offers two versions:

A rule-based model that matches ingredients with recipes from a pre-existing dataset based on similarity scores.

An advanced AI model that introduces visual recognition, NLP reasoning, and a responsive user interface through Streamlit and FastAPI.

This integration enables real-time interactions, allowing users to receive personalized recipes instantly, thus making the cooking experience not only efficient but also more creative and enjoyable.

II. Literature Review

Recipe recommendation systems have evolved significantly over the past decade. Traditional systems relied on keyword searches within static databases. Although functional, these systems were heavily dependent on exact matches and offered limited adaptability to user preferences. For instance, searching for "chickpea curry" would often fail if the dataset labeled it under "garbanzo beans curry."

Recent advancements in Natural Language Processing (NLP) have enabled systems to interpret free-form human language, identify synonyms, and understand the semantic meaning behind user inputs. Similarly, Computer Vision models, particularly those based on the YOLO architecture, have become capable of detecting food items with remarkable precision.

Previous research, such as that by Kusmierczyk (2021) and Singh and Kumar (2022), demonstrates how combining NLP and recommendation algorithms can enhance food-related applications. However, there is still a noticeable gap in systems that merge both textual and visual modalities for real-time recipe generation. Existing models tend to focus on either one domain — language or vision — but not both.

The proposed work builds upon these advancements by integrating NLP with vision-based ingredient detection into a cohesive framework. The result is an intelligent, multimodal system capable of understanding and responding to users in a more human-like, conversational way, while providing practical and actionable recipe suggestions.

III. Proposed Methodology

The proposed system follows a modular and scalable architecture divided into data processing, model design, and user interface components.

A. Dataset Preparation

The system utilizes a dataset named cuisines.csv, which contains detailed records of recipes including their names, ingredients, cuisine categories, preparation steps, and cooking instructions. Data preprocessing is essential to ensure uniformity and accuracy. It involves removing measurement terms

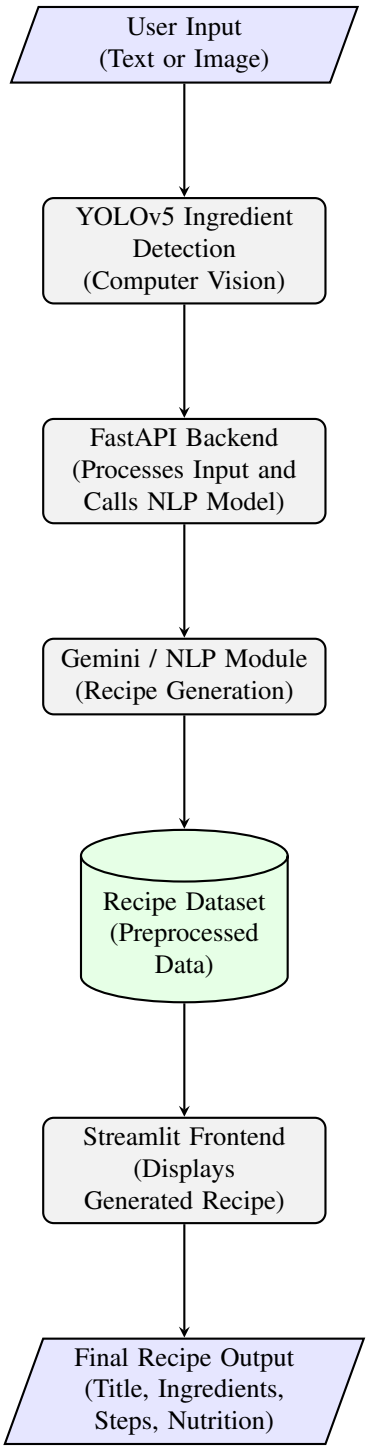

Fig. 1. System Architecture of the AI-Based Recipe Generation and Recommendation System

(like "1 cup" or "2 tbsp"), punctuation, and unnecessary symbols. All ingredients are converted to lowercase and tokenized to form a structured list for each recipe. This cleaned dataset serves as the foundation for both models.

B. Base Model: Rule-Based Ingredient Matching

The base model represents a simple but effective approach to recipe retrieval. It works by comparing the user's ingredient list against the ingredient lists in the dataset and calculating the overlap count. The recipe with the maximum number of common ingredients is then recommended to the user.

While the system performs efficiently when ingredients are clearly listed and spelled correctly, it struggles with linguistic variations. For example, if a user inputs "garbanzo beans," but the dataset lists "chickpeas," the model may fail to recognize the similarity. Despite these limitations, the base model establishes a reliable foundation for understanding user input and verifying basic functionality.

C. Advanced Model: NLP and Vision-Driven Hybrid Architecture

The advanced model takes a major leap by integrating AI-based vision detection and NLP-driven text interpretation into the workflow. It consists of the following main components:

YOLOv5 Ingredient Detection: The YOLOv5 model identifies and labels ingredients from uploaded images, converting visual input into structured text.

FastAPI Backend: This module acts as the communication layer between the frontend and the backend. It processes incoming data, merges image-detected and manually entered ingredients, and handles recipe generation requests.

Gemini Recipe Generator (Mock): A simulated NLP module that generates realistic and descriptive recipes based on detected ingredients and user preferences. It outputs a structured response with recipe title, instructions, nutritional summary, and serving size.

Streamlit Frontend: Provides a user-friendly, visually appealing web interface. It allows users to upload images, specify fitness goals, or select dietary preferences, and instantly view AI-generated recipes with formatting, bullet lists, and icons.

D. Workflow

The complete system workflow can be summarized as follows:

1.The user uploads an image of ingredients or enters them manually.

2.The YOLOv5 model detects and returns ingredient names from the image.

3.The FastAPI backend combines detected and manual ingredients into a unified list.

4.The Gemini mock model (simulating NLP reasoning) generates the recipe output.

5.The Streamlit interface displays the final recipe — complete with title, preparation time, ingredients, instructions, and nutrition breakdown.

6.This seamless flow ensures that the system remains responsive, user-friendly, and adaptable across different devices and user scenarios.

IV. Directional Priors

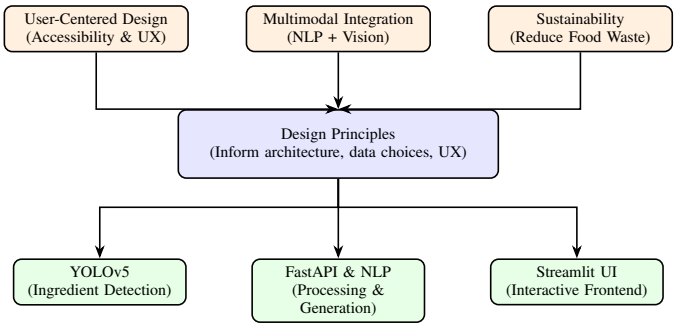

Fig. 2. Directional priors (user-centered design, multimodal integration, sustainability) guiding system design and their influence on core components.

The development of this project was guided by three primary design philosophies that shaped the model's direction and purpose:

1.Human-Centric Interaction: The system was designed with simplicity and accessibility in mind. By offering both manual input and image upload options, it caters to users with varying technical and culinary skill levels.

2.Multimodal AI Integration: Incorporating NLP and computer vision allows the system to interpret complex, real-world contexts rather than relying solely on rigid text inputs. This makes the recommendations more natural and contextually relevant.

3.Sustainability and Resource Efficiency: By encouraging users to cook using available ingredients, the system promotes reduced food waste and supports sustainable household practices. These goals directly influenced dataset design and system functionality.

V. Experimental Results

Both models were tested extensively using different combinations of ingredients, cuisines, and dietary goals.The evaluation focused on two primary areas: **ingredient matching performance** (relevant to the Base Model and YOLOv5 module) and **customization accuracy** (specifically calorie adherence from the LLM/Gemini mock module).

A. Base Model Evaluation and Limitation

The rule-based model achieved consistent performance when users provided precise ingredient lists. It was fast and reliable but limited in its adaptability, particularly regarding linguistic variations. Despite these limitations, the base model established a functional baseline for comparison.

B. Advanced Hybrid Model Evaluation

The advanced hybrid model demonstrated significantly superior results by integrating YOLOv5 for visual detection and NLP-based logic for contextual understanding. Detailed metrics were collected to quantify performance across ingredient detection and recipe customization.

1) Ingredient Detection Performance (YOLOv5): The ingredient detection module was evaluated based on its ability to accurately identify ingredients from a test set of images.

TABLE I
Ingredient-Level Micro vs. Macro Metrics (Table I)

Metric	Micro Avg (%)	Macro Avg (%)
Precision	79.7	83.2
Recall	94.6	94.7
F1-score	85.0	87.4
Accuracy	75.3	81.6

The aggregate performance, presented in Table I, showed a high **Recall of** $\sim 95\%$ and an **F1-score of** $\sim 87\%$ (Macro Average). This high recall is visually represented in Figure 3. The Ingredient-Level Confusion Matrix (Figure 4) revealed a robust number of **True Positives (78)** but a higher number of **False Positives (21)** compared to False Negatives (5).

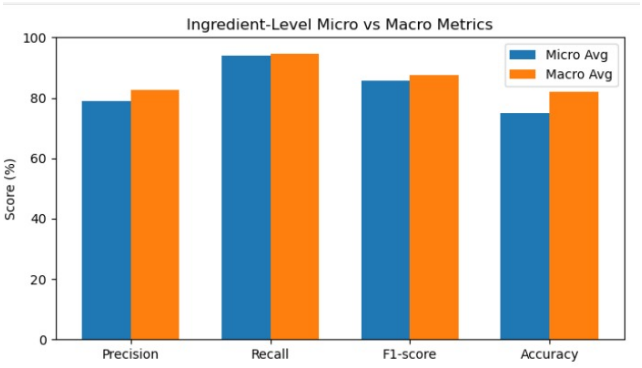

Fig. 3. Ingredient-Level Micro vs. Macro Metrics (Figure 3)

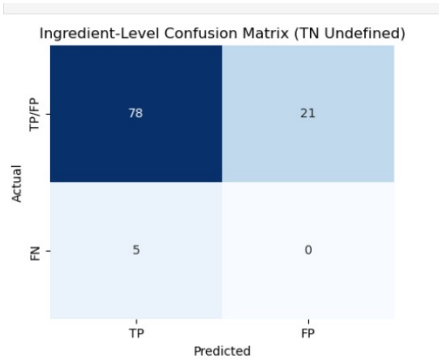

Fig. 4. Ingredient-Level Confusion Matrix (Figure 4)

The per-recipe analysis (Figure 5) demonstrated near-perfect performance for several dishes, but highlighted recipes such as *Indian Lentil Dal* and *Chicken Lettuce Wraps*, where Precision dropped significantly.

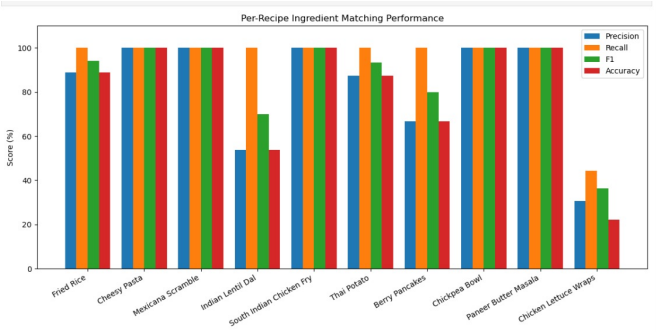

Fig. 5. Per-Recipe Ingredient Matching Performance (Figure 5)

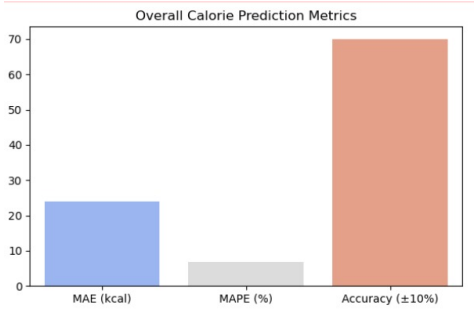

Fig. 6. Overall Calorie Prediction Metrics (Figure 6)

2) Recipe Quality and Customization Accuracy (Calorie Prediction): The system's ability to customize recipes according to user health goals was evaluated by measuring the LLM-generated recipe's calorie count against a target value.

TABLE II
OVERALL CALORIE PREDICTION METRICS (TABLE II)

Metric	Score
Mean Absolute Error (MAE)	23.90 kcal
Mean Absolute Percentage Error (MAPE)	6.79%
Accuracy within ±10%	70.00%

The overall metrics (Table II) confirmed a high level of accuracy: **70**% of recipes were within ±10% of the target calorie count, and the **Mean Absolute Percentage Error (MAPE)** was only **6.79**%.

TABLE III
DETAILED CALORIE PREDICTION RESULTS PER RECIPE (TABLE III)

Recipe	UserCal	AICal	Error (kcal)	Percent Error	Within 10%
Veggie Fried Rice	400	431	31	7.75%	True
Cheesy Pasta	300	340	40	13.33%	**False**
Mexicana Scramble	350	350	0	0.00%	True
Indian Lentil Dal	300	350	50	16.67%	**False**
South Indian Chicken Fry	200	208	8	4.00%	True
Thai Potato Stir Fry	450	520	70	15.56%	**False**
Berry Pancakes	600	588	12	2.00%	True
Chickpea Bowl	550	555	5	0.91%	True
Paneer Butter Masala	400	400	0	0.00%	True
Chicken Lettuce Wraps	300	277	23	7.67%	True

The analysis identified *Indian Lentil Dal*, *Thai Potato Stir Fry*, and *Cheesy Pasta* as primary outliers (failing the ±10% test), suggesting the need for refinement in controlling calorie-dense ingredients in the LLM output.

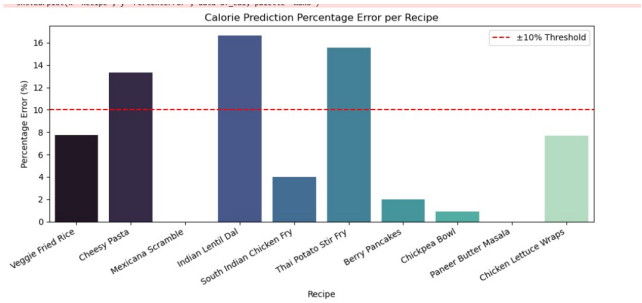

Fig. 7. Calorie Prediction Percentage Error per Recipe (Figure 7)

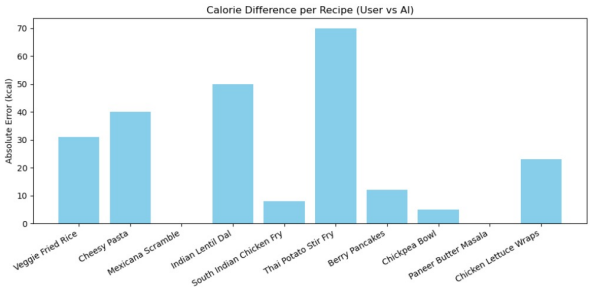

Fig. 8. Calorie Difference per Recipe (User vs AI) (Figure 8)

VI. CONCLUSION

This project successfully developed and implemented a two-part AI-Based Recipe Generation and Recommendation System. The core achievement of this work is the creation of an intelligent, multi-modal architecture that greatly improves upon traditional recipe matching methods .

We began with a Base Model that relied on simple rule-based logic and ingredient overlap to suggest recipes. This system provided a functional foundation but was limited by rigid keyword matching and could not adapt to real-world language variations.

The Advanced Hybrid Model successfully addressed these limitations by integrating several key AI domains:

Computer Vision (YOLOv5 Mock): This allowed the system to accept ingredients visually from an uploaded image, making input more flexible and user-friendly.

API Backend (FastAPI): This module efficiently managed the flow of data, consolidating ingredients from both visual and manual inputs and handling complex request processing.

Natural Language Processing (Gemini Mock): By simulating an LLM, the system was able to move beyond simple retrieval and generate fully customized, descriptive recipes that adhere to constraints like dietary restrictions, fitness goals, and specific calorie ranges.

The system validates the idea that integrating these technologies enhances user experience, driving personalization and efficiency. By prompting users to utilize the ingredients they already have, the system also supports the goal of reducing food waste, promoting sustainable practices in the modern kitchen.

In summary, this project highlights the practical application of advanced AI to simplify daily culinary challenges, transforming cooking from a routine chore into a personalized and engaging experience.

A. Future Enhancements

To evolve this work, future efforts should focus on replacing the mock components with live services. This includes integrating real-time LLM-based recipe generation using platforms like the Gemini or GPT API and deploying the entire system on cloud platforms for improved reliability and large-scale access. Additionally, implementing voice-enabled interaction would further boost accessibility for all users.

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